Minseong Kim

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Education

- **Inha University** *Mar. 2021 – Feb. 2027 (Expected)*
  B.S. in Data Science; Double Major in Artificial Intelligence Engineering Incheon, Republic of Korea
  - Overall GPA: 4.25/4.50 (Honor Student, all semesters)
  - Major GPA: 4.29/4.50
  - Credits: 150 (including current semester; 130 required)

Research Interests

- Trustworthy AI
- Multimodal Representation Learning
- Statistical Machine Learning

Publications & Manuscripts

- **Minseong Kim***, Junhee Kim*, Seongil Jo, and Jaeoh Kim. "SAM-BO: Perturbation-Radius Selection via an Empirical PAC-Bayes Proxy." *To be submitted.* (*Equal contribution)
- Jeongwan Park, Yoonsuk Jang, **Minseong Kim**, Taeyoung Park, Seongil Jo, Doowon Choi, and Jaeoh Kim (2026). "Class-Aware Topology-Guided Graph Attention for Large-Scale Skeleton-Based Firefighter Action Recognition." *Applied Soft Computing*, 116098. [Paper]
- Minseo Kim, Youngdam Choi, **Minseong Kim**, Yunsang Kim, Yongyoon Cho, Jong Sung Won, Man Young Lee, Jaeoh Kim, and Youngho Jin (2026). "PDA-Based Interphase Engineering with CNT/GO Hybrid Nanofillers for Balanced Aramid/Epoxy Composites." *Preprint.* [SSRN]

Intellectual Property

- **SAM-BO: Bayesian Optimization for Sharpness-Aware Minimization** *Jun. 2026*
  Korea Copyright Commission (KCC)
  - Software Copyright Registration, Reg. No. C-2026-027538
  - Registered the implementation of the SAM-BO radius-selection framework accompanying the manuscript above

Research Experience

- **PCR Lab, Inha University** *Jan. 2025 – Present*
  Undergraduate Researcher Advisor: Prof. Jaeoh Kim
  - Conduct research on generalization theory and multimodal representation learning
  - Co-led PAC-Bayes-guided perturbation-radius selection for SAM (SAM-BO) as co-first author, with registered software
  - Implemented models and ran experiments for class-aware graph attention in skeleton-based action recognition

- **MIND Lab, Inha University** *Sep. 2024 – Dec. 2024*
  Undergraduate Researcher Advisor: Prof. Doowon Choi
  - Studied causal inference from observational data, building a foundation for research on trustworthy AI

Projects

- **Intelligent GOP Operational Support Framework Based on K-AIP** *2025 – 2026*
  Korea Research Institute for Defense Technology Planning and Advancement (KRIT)
  – Researched ontology-guided spatio-temporal scene graphs within a multimodal fusion pipeline that turns surveillance video and text logs into structured situational reports

- **AI-Based Environment-Aware Sensing Optimization for Personal CBRN Contamination Detection** *2026*
  Agency for Defense Development (ADD)
  – Developed an ensemble classifier robust to sensor noise, drift, and signal loss for multi-sensor gas classification, and delivered the final software

- **Vision-Based Object Detection and Tracking for UAVs** *2025*
  Poongsan Corporation
  – Fine-tuned YOLO on simulator-collected imagery and built a multi-to-single object tracking pipeline with improved position estimation

Awards & Scholarships

- **Excellence Award,** 2026 Inha AI Challenge, DACON *2026*
  – Awarded in a competition on world-model-based video generation for predicting future robot motion from current observations and action sequences

- **Academic and Extracurricular Excellence Award,** College of AI Convergence, Inha University *2026*
  – Awarded for outstanding academic performance and student activities

- **Silla Scholarship,** Silla Scholarship Foundation *2024 – 2026*
  – Full tuition scholarship for six consecutive semesters

- **Unhae Scholarship,** Unhae Scholarship Foundation *2024 – 2025*
  – Living-expense stipend awarded for two consecutive academic years

- **Seoil Petroleum Scholarship,** Inha University Alumni Association *2025*
  – Scholarship funded by the alumni association

- **Academic Excellence Scholarship,** Inha University *2021*
  – One-semester full tuition scholarship for outstanding academic performance

- **Pohang Scholarship,** Pohang Scholarship Foundation *2021*
  – One-semester full tuition scholarship

Additional Experience

- **Student Council Member,** Unhae Scholarship Foundation *2025*
  – Planned and ran activities for fellow scholarship recipients

- **Peer Mentor,** Department of Data Science, Inha University *2024*
  – Mentored underclassmen in core data science coursework

- **Military Service,** Republic of Korea Army *2022 – 2023*
  – Driver, Protocol Division, Office of the Minister of National Defense